

Farmer Interview Report

Name: Ramasubbu

Place: Vembathi, Tamil Nadu, India

Address: Vembathi, Aapakadal, Tamil Nadu 638315, India

Interview Questions and Answers

1. What crops do you cultivate?

Answer: We cultivate sugarcane, turmeric, and banana.

2. How many years have you been farming?

Answer: I have been farming for 25 years.

3. If you need to apply a particular medicine or treatment to the crop, how do you know what the problem is and which medicine should be applied?

Answer: We mainly depend on our farming experience. We have been farming for many years and use our experience to identify the problems.

4. Do you usually ask at agricultural shops for advice?

Answer: Yes. If there is any disease or problem in the crop, we observe it and ask at the agricultural shop. They tell us what treatment should be given.

5. Have you ever given your soil for testing?

Answer: No, I have not given the soil for testing.

6. If you are not sure which medicine to apply and later notice something unusual, such as changes in the leaves, how do you identify the problem and what is your next step?

Answer: We ask agricultural experts or the agricultural shop. Based on their advice, we take the necessary action. We mainly supply our sugarcane to the factory. If there is any complaint or problem, we inform them and they come and inspect it. They provide the appropriate solution, and we follow their instructions.

7. How much water does sugarcane require?

Answer: If the soil is clayey, we irrigate once every 10 days. If it is red soil, we irrigate approximately once every 6 or 7 days.

8. Are you using drip irrigation?

Answer: No, we have not followed drip irrigation so far.

9. Do you normally irrigate the field manually using a motor?

Answer: Yes. We follow conventional/manual irrigation practices.

10. If our project could be brought directly to your mobile phone, how useful would it be for you?

Answer: Since it is a new method, I would like to try it and see how useful it is.

11. If the system automatically determines when water is required and irrigates the crop without you having to go to the field, how would you respond to it?

Answer: Yes, that would be useful.

12. If the system could also automatically determine the required fertilizer and apply the required quantity, to what extent would you be willing to use it?

Answer: We would definitely cooperate. If the method works well and is practical, we will definitely support and cooperate with it.

13. Normally, how much do you spend on cultivation? Is the cost different for each crop?

Answer: For sugarcane, the total cultivation cost is around ■40,000 per acre. For banana, it costs more than ■1,00,000 per acre.

14. Does that include all the expenses?

Answer: Yes, it is the overall cost, including all expenses.

15. After spending that much, how much yield do you usually get? What kind of returns do you receive?

Answer: If we supply sugarcane to the factory, we can expect around 45–50 tonnes of yield. For banana, the returns depend on the prevailing market price. If the price is high, we get better returns.

16. Have you ever used any application or technology before that allows you to switch something ON or OFF from your home, like the system we are proposing?

Answer: No. I have not used anything like that technically before.

Farmer Interview Report

Name: Rajagopalan

Place: Vembathi

Address: Aapakudal, Bhavani, Erode district, TamilNadu-638315

Interview Questions and Answers

1. What is your name, and how many years have you been involved in agriculture?

Answer: The farmer's name is Rajagopalan. He has been involved in agriculture for approximately 20–25 years.

2. What crops have you planted?

Answer: He has cultivated sugarcane, banana, and turmeric. Among these, sugarcane is the crop he has cultivated mostly.

3. How many months does it take for banana cultivation from planting to harvesting?

Answer: The duration depends on the banana variety. For example, the Kathali variety produces a bunch around the 6th or 7th month and becomes ready for harvest in approximately 10 months. The entire field can be harvested within about one year.

4. How frequently do you provide water to the crops?

Answer: After planting, the crops are generally irrigated once every 6–8 days, depending on the requirement.

5. Do you use drip irrigation or manual/flood irrigation?

Answer: He does not use drip irrigation. The farm mainly uses flood irrigation/regular water irrigation.

6. How much does irrigation cost?

Answer: He does not maintain a separate account for irrigation costs because the family members themselves carry out most of the agricultural work. Outside workers are hired only for major tasks.

7. How do you decide which pesticides or fertilizers to apply?

Answer: An agricultural officer visits the farm, and the farmer consults the officer about which pesticides or fertilizers should be applied.

8. Have you tested your soil in a laboratory?

Answer: Yes. The soil was tested in a laboratory more than 10 years ago. A soil sample was collected and tested, but they have not conducted another soil test recently on their own.

9. Does the water supply change according to the season?

Answer: The farm mainly depends on well-water irrigation and sometimes river irrigation. Based on the available water, they generally irrigate the crops once a week.

10. Would a system that allows you to control the water pump through your mobile phone be useful?

Answer: According to the farmer, such systems are mainly useful for farms that use large motors such as 20 HP or 50 HP motors, where the farmer may need to switch the motor on or off from a distance. Since his farm is nearby and he can operate the system directly, he does not currently feel a strong need for remote control.

11. Have you noticed differences in crop yield within the same field?

Answer: Yes. He has observed that some areas of the same field produce better yields than others. This mainly depends on the nature and quality of the soil. Areas with rocky or gravelly soil generally have lower yields.

12. How do you improve the yield in areas where the soil is not good?

Answer: He manages the water supply to some extent and applies additional fertilizers where required, especially when there is a salt-content problem. Another method is to add manure to the soil and prepare it properly to improve the soil condition.

13. How many acres of land do you own?

Answer: The farmer owns approximately 4 acres of agricultural land.

Farmer Interview Report

Name: Selvaraj

Place: T. N Palayam

Address: T. N Palayam, Gobichettipalayam, Erode district, Tamil Nadu - 638506

Interview Questions and Answers

1. What is your name, and how many years have you been involved in agriculture?

Answer: The farmer's name is Selvaraj. He has been involved in agriculture for approximately 25 years.

2. What crops have you planted?

Answer: He is currently cultivating banana (plantain). Previously, he cultivated turmeric, but after harvesting it, he transitioned the field entirely to banana.

3. Why did you transition from turmeric to banana cultivation?

Answer: The primary reason for shifting crops was a severe shortage of agricultural labor. Turmeric requires a significant amount of manual labor to cultivate and harvest, whereas banana cultivation requires considerably fewer laborers and lower overall labor costs.

4. How long does banana cultivation take from planting to harvest?

Answer: Banana cultivation takes approximately 10 months from the initial planting stage to full harvest readiness.

5. Do you use drip irrigation or manual/flood irrigation, and what is your water source?

Answer: He uses drip irrigation supplied by a borewell.

6. How much does the drip irrigation system cost to install?

Answer: Installing the drip irrigation tubing/hose network alone costs approximately ₹70,000 per acre. If the borewell motor, pipeline, and main electrical line connections are included, the total infrastructure investment reaches roughly ₹2,00,000.

7. How many acres of agricultural land do you own?

Answer: The farmer owns a total of 4 acres of land.

8. Have you tested your soil or water in a laboratory?

Answer: Yes. He tested both the soil and the borewell water in an agricultural laboratory about six years ago; however, he does not remember the exact numerical test parameters.

9. How do you decide which pesticides or fertilizers to apply?

Answer: Representatives and field officers from private agrochemical firms and government agricultural extension departments visit the field approximately two to three times a year to inspect the crop and prescribe the necessary fertilizers and treatments.

10. What fertilizers do you commonly apply to the banana crop?

Answer: He primarily applies Potash, Urea, Complex fertilizer, and DAP (Di-ammonium Phosphate), with Potash being the most heavily applied nutrient.

11. Does irrigation demand vary across seasons?

Answer: Yes. During the rainy season, irrigation needs drop significantly. Water demand peaks during the hot summer months, although product market prices vary based on seasonal supply rather than rainfall alone.

12. Have you noticed differences in crop yield within the same field?

Answer: Yes. Variations in growth occur across different sections of the field. He attributes these differences primarily to uneven soil characteristics and slight quality variations in the planting material/seedlings used.

13. Do you practice intercropping alongside banana cultivation?

Answer: No. While it is possible to grow short-duration intercrops during the first three months of planting, he avoids it because intercrops deplete nutrients and reduce the final banana yield. The combined revenue usually ends up the same or less, making monocropping the simpler and better option.

14. Would a mobile-based automated motor starter system be suitable for your farm?

Answer: While he considers remote automated irrigation systems useful in principle, it is not practical for his current water situation. Because his borewell water yield fluctuates and runs low, he has to run the motor intermittently in short cycles rather than continuously, which requires manual supervision.

15. Would an automated soil-sensing and fertilizer advisory application be useful to you?

Answer: Yes. Having direct access to localized soil data and automated fertilizer recommendations through a mobile device would be very helpful, as it would reduce dependence on visiting agents and make routine crop nutrition management much easier.